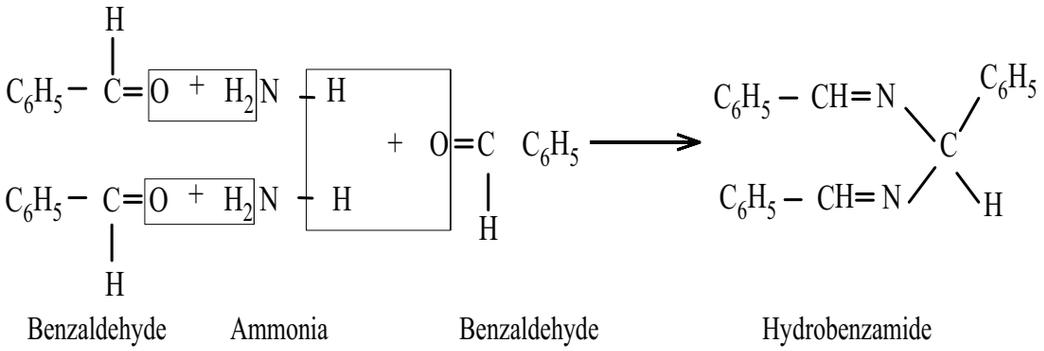
**iv)** Benzaldehyde form a complex condensation product with ammonia.

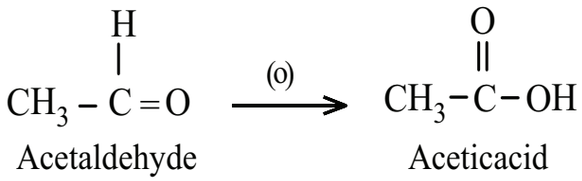



## B) Oxidation of aldehydes and ketones

### a) Oxidation of aldehydes

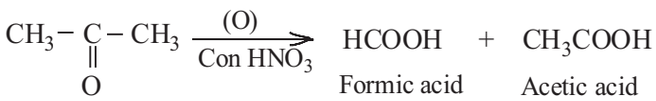
Aldehydes are easily oxidised to carboxylic acid containing the same number of carbon atom, as in parent aldehyde. The common oxidising agents are acidified $K_2Cr_2O_7$, acidic or alkaline $KMnO_4$ or chromic oxide.

**Example**

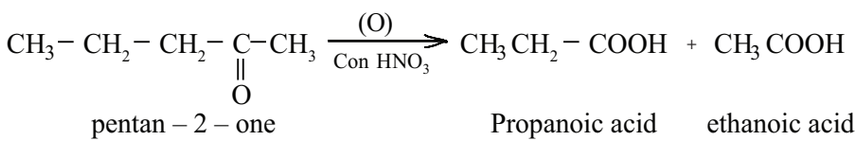



### b) Oxidation of ketone

Ketones are not easily oxidised. Under drastic condition or with powerful oxidising agent like Con.$HNO_3$, $H^+/KMnO_4$, $H^+/K_2Cr_2O_7$, cleavage of carbon-carbon bond takes place to give a mixture of carboxylic acids having less number of carbon atom than the parent ketone.




The oxidation of unsymmetrical ketones is governed by **Popoff's rule**. It states that during the oxidation of an unsymmetrical ketone, a (C–CO) bond is cleaved in such a way that the keto group stays with the smaller alkyl group.






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